Jitesh **Jain**

Ph.D. Student | Computer Science | Georgia Tech

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Education

Georgia Tech — *Atlanta (GA)*

Ph.D. Student in Computer Science — *Since Fall 2023; Expected Fall 2026*

- Advisor: Prof. Humphrey Shi
- CGPA: 4.0/4.0

Indian Institute of Technology Roorkee — *Roorkee, India*

B.Tech. in Computer Science and Engineering — *2019 - 2023*

- CGPA: 9.0/10

Experience

PRIOR @ Allen AI — *Seattle (WA)*

Research Intern | Mentor: Dr. Christopher Clark — *May 2025 - Dec 2025*

- *Research:* Training Long Video Understanding Agents with Reinforcement Learning.

Microsoft Research — *Redmond (WA)*

Research Intern | Mentor: Dr. Jianwei Yang — *May 2024 - Dec 2024*

- *Research:* Improving Visual Perception in MLLMs with Embedding Distillation.

SHI Labs @ UIUC & U of Oregon — *Eugene (OR)*

Research Intern | Mentor: Prof. Humphrey Shi — *November 2020 - June 2023*

- *Research:* Unifying Image Segmentation with a Single Architecture, Model and Dataset.

Publications

For a complete list of my research works, please lookup my **google-scholar**.

SAGE: Training Smart Any-Horizon Agents for Long Video Reasoning with Reinforcement Learning — *arXiv 2025*

Jitesh Jain, Jialuo Li, Zixian Ma, Jieyu Zhang, Chris Dongjoo Kim, Sangho Lee, Rohun Tripathi, Tanmay Gupta, Christopher Clark, Humphrey Shi — *PRIOR @Allen AI*

- **[Paper] [GitHub] [Project Page]**

Elevating Visual Perception in Multimodal LLMs with Visual Embedding Distillation — *NeurIPS 2025*

Jitesh Jain, Zhengyuan Yang, Humphrey Shi, Jianfeng Gao, Jianwei Yang — *Microsoft Research, Redmond*

- **[Paper] [GitHub] [Project Page]**

AUGUSTUS: An LLM-Driven Multimodal Agent System with Contextualized User Memory — *LAW @NeurIPS 2025*

Jitesh Jain*, Shubham Maheshwari*, Ning Yu, Wen-mei Hwu, Humphrey Shi — *SHI Labs @ GaTech*

- **[Paper] [GitHub][Project Page]**

VCoder: Versatile Vision Encoders for Multimodal Large Language Models — *CVPR 2024*

Jitesh Jain, Jianwei Yang, Humphrey Shi — *SHI Labs @ GaTech*

- **[Paper] [GitHub] [Project Page]**

OneFormer: One Transformer to Rule Universal Image Segmentation — *CVPR 2023*

Jitesh Jain, Jiachen Li*, MangTik Chiu*, Ali Hassani, Nikita Orlov, Humphrey Shi — *SHI Labs @ UIUC & UO*

- **[Paper] [GitHub] [Project Page]**

Benchmarking Object Detectors with COCO: A New Path Forward — *ECCV 2024*

Shweta Singh*, Aayan Yadav*, **Jitesh Jain**, Humphrey Shi, Justin Johnson, Karan Desai — *SHI Labs @ GaTech*

- **[Paper] [GitHub] [Project Page]**

Keys to Better Image Inpainting: Structure and Texture Go Hand in Hand — *WACV 2023*

Jitesh Jain*, Yuqian Zhou*, Ning Yu, Humphrey Shi — *SHI Labs @ UIUC & UO*

[Paper] [GitHub] [Project Page]

References

Prof. Humphrey Shi *United States*

Associate Professor @Georgia Tech

- **Email**: shi@gatech.edu

Dr. Jianwei Yang *United States*

Research Scientist @Meta SuperIntelligence Lab

- **Email**: jw2.yang@gmail.com

Dr. Christopher Clark *United States*

Research Scientist @Allen AI

- **Email**: chrisc@allenai.org

Dr. Ning Yu *United States*

Research Scientist @Netflix Eyeline Studios

- **Email**: ningyu.hust@gmail.com

Honours & Achievements

2025 **NeurIPS Scholar Award**, Awarded a full conference registration and a 7-night San Diego hotel stay to attend NeurIPS 2025.

2022 **Bronze Medalist**, Awarded the Bronze Medal under Bosch's Age and Gender Detection Track at Inter IIT Tech Meet 10.0. The competition is held every year among all the IITs.

2019 **All India Rank - 252**, Joint Entrance Exam (JEE) Advanced is attempted by more than 230,000 students across India for admission into the IITs.

2019 **All India Rank - 454**, Joint Entrance Exam (JEE) Main is attempted by more than 1,000,000 students across India to gain eligibility for attempting JEE Advanced.

2018 **KVPY Fellowship Holder**, KVPY is a highly prestigious National Fellowship awarded by the Indian Institute of Science to students showing high aptitude in research.

2017 **NTSE Scholarship Awardee**, NTSE is a national level examination for the secondary school level to identify and recognize students with high intellect and skills.

Extracurricular Activity

Vision and Language Group, IIT Roorkee *Roorkee, India*

Co-President *January 2020 - May 2023*

- Co-President of VLG IITR, a student group that promotes Deep Learning research culture at IITR by discussing relevant research papers, organizing workshops and working on projects. **[Website]**

MDG Space, IIT Roorkee *Roorkee, India*

Project Manager *January 2020 - May 2023*

- Project Manager at MDG Space in IIT Roorkee, involved in developing and managing mobile applications in Flutter. **[Website]**

Student Mentorship Program, IIT Roorkee *Roorkee, India*

Student Mentor *October 2021 - May 2022*

- Mentored freshers at IITR for their smooth transition to campus life, motivating their academic and co-curricular endeavors. **[Certificate]**